

Radon Mitigation Is Not the End of the Evidence

From Initial Measurement to Verified Environmental Consequence

INDEPENDENT TECHNICAL PRE-EXAMINATION | REFERENCE v1.0 | 10 OCTOBER 2026

PRE-EXAMINATION | BASELINE NOT FROZEN | NO FORMAL DISPOSITION

Independent educational architecture demonstration. No endorsement, affiliation, approval, certification, or operational integration is claimed from the Indoor Environments Association (AARST), RadonAway, EPA, or any other institution. All case facts and sample records are hypothetical.

Governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Purpose: show how a radon-related environmental observation can remain traceable through interpretation, mitigation, post-intervention verification, and future changed conditions. This reference is not a radon testing protocol, mitigation design, exposure assessment, or substitute for applicable standards and qualified professionals.

1. Scope and examination boundary

This examination considers a hypothetical occupied building with an elevated radon test result, a proposed mitigation intervention, and a subsequent verification measurement. It does not assign a universal action threshold, determine health risk for an individual, or authorize work. Applicable jurisdictional requirements, test protocols, qualifications, and building-specific conditions must be independently established.

TA-14 does not replace professional testing, mitigation standards, code officials, owners, or regulators. Its proposed contribution is a separable evidence-and-authority examination of the decisions made around those practices.

2. The eight-anchor chain

Anchor	Required distinction
REALITY	Actual environmental and building conditions, not merely dashboard representations.
RECORD	Attributable original measurement and contextual metadata.
CONTINUITY	Unbroken provenance across people, systems, versions, and transfers.
ADMISSIBILITY	Fitness of evidence for the specific decision under examination.
BINDING	Connection of evidence to the specific building, scope, responsible party, and authority.
COMMIT	Documented bounded decision: ALLOW, HOLD, DENY, or ESCALATE.
EXECUTION	Intervention performed within locally established authority.
OUTCOME	Verified result, limitations, and conditions for re-examination.

3. Illustrative record package

A defensible case package might contain building/zone identity; sampling location and dates; measurement method, duration, device or laboratory identity and quality records; raw results and units; occupant/use context; professional interpretation; mitigation scope and approvals; installation and commissioning evidence; follow-up test details; and subsequent changes. These are illustrative record categories, not a replacement for prescribed testing requirements.

4. Eight-image teaching sequence

01 - The Invisible Condition What was measured, where, when, and by what method? A result is an observation with limits, not a permanent characterization.

02 - The Original Record Retain original results and metadata; distinguish raw readings from reports, averages, corrections, and interpretations.

03 - Evidence Continuity Track custody, transformations, instrument or lab records, file versions, and any missing context.

04 - The Mitigation Decision Establish who may propose, approve, perform, and accept work in the relevant jurisdiction and contractual setting.

05 - Installation Is Not Verification Document as-built work and checks, but do not equate installed equipment with demonstrated environmental performance.

06 - The Post-Mitigation Test Use applicable qualified procedures to assess the relevant outcome, retaining the actual result and its limitations.

07 - Changed Conditions Reassess when building use, ventilation, pressure relationships, equipment, occupancy, or other relevant conditions change.

08 - Authorized Environmental Consequence State only the bounded conclusion supported by evidence and standing at that time; preserve uncertainty and future obligations.

5. Hypothetical disposition example

Scenario: An initial radon result prompts a proposed mitigation installation. Work completion is documented, but the post-mitigation measurement record is missing or cannot be attributed to the treated zone.

Illustrative disposition: HOLD. The available evidence may support that work was performed; it does not yet support a verified post-mitigation outcome. The responsible authority should obtain appropriate verification and re-examine the claim. This is a teaching example, not an actual TA-14 ruling.

6. Responsibility boundaries

Role	Boundary to establish
Testing professional / laboratory	Measurement method, validity, attribution, reporting.
Mitigation professional	Design and installation scope, qualifications, work records.
Owner / authorized representative	Approvals, access, acceptance, ongoing responsibilities.
Authority having jurisdiction	Applicable legal and regulatory requirements where relevant.

Role	Boundary to establish
TA-14 examination layer	Evidence continuity and decision-boundary assessment; no assumption of statutory authority.

7. Changed-context revalidation

A satisfactory result at one time does not prove perpetual conditions. Any future claim must account for the relevant time, building configuration, environmental conditions, method, and intervening changes. Preserve both original and subsequent measurements; never overwrite the original record with a revised interpretation.

8. Examination protocol

Step A - Identify consequence: specify exactly what is proposed: installation approval, operational acceptance, a statement of mitigation effectiveness, or another bounded claim.

Step B - Freeze candidate evidence: inventory sources, versions, methods, dates, provenance, and known gaps. A frozen baseline is required before a formal examination; none is frozen for this educational reference.

Step C - Establish authority and standing: identify relevant owners, qualified professionals, and applicable regulatory or contractual requirements. Registration with TA-14, where applicable to its examination process, does not confer regulatory authority.

Step D - Evaluate: examine each of the eight anchors and assign a bounded ALLOW / HOLD / DENY / ESCALATE disposition only within a properly established examination.

Step E - Preserve outcome: retain the decision rationale, conditions, evidence identifiers, dissent or uncertainty, and revalidation triggers.

9. What this architecture does not claim

It does not assert that an isolated test establishes an individual health outcome; that continuous sensing replaces prescribed radon measurement methods; that a mitigator can self-grant legal authority; or that TA-14 can issue radon certification or regulatory clearance. It also does not claim that a radon-specific deployment or integration has been completed.

10. Candidate institutional demonstration

A bounded pilot could use a synthetic or appropriately permissioned de-identified case to examine handoffs from measurement to mitigation to verification. Success criteria would include attributable source records, traceable transformations, documented decision rights, explicit HOLD behavior for missing evidence, and a reproducible final evidence package. Any real-world pilot would require a separately agreed scope and governance registration before formal TA-14 examination.

Reference and verification note

Relevant external frameworks may include current EPA radon guidance, ANSI/AARST measurement and mitigation standards, and applicable state or local requirements. Exact editions, jurisdictional applicability, thresholds, and professional qualifications must be verified for each actual project. No external standard is reproduced or certified in this document.

TA-14 AUTHORITY GOVERNANCE INSTITUTION | Independent technical pre-examination | v1.0